

STA141B Final Project

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1 Abstract

This study analyzes federal job postings from USAJOBS, the official U.S. government website for finding and applying to federal jobs, to examine geographic distribution patterns, cost-of-living-adjusted salaries, and the requirements. We collected about 8,000 job postings through the USAJOBS API, processed location data to map jobs to an area, and adjusted salaries using Regional Price Parity data. Our analysis shows significant geographic concentration in coastal states and substantial variation in real purchasing power across regions. We developed an interactive job search tool that filters positions by education level and location, calculating both high-paying and nearest opportunities. The methodology demonstrates the importance of cost-of-living adjustments when comparing federal job opportunities across different geographic regions.

2 Introduction

Federal employment opportunities vary substantially across geographic regions due to differences in local labor demand, agency presence, and regional cost of living. A normal salary, for example, of \$80,000 in San Francisco provides less purchasing power than the same salary in Fargo, North Dakota. For job seekers, understanding these geographic and economic differences is essential for making informed decisions about where to apply and whether relocation is worthwhile, especially for graduates. In addition, federal job postings vary widely in education requirements, and applicants benefit from tools that identify positions that align with their qualifications and preferred locations.

Our project addresses three main research questions:

1. Geographic Distribution: What is the geographic distribution and patterns of federal job postings by state?
2. Education Visualization: How can education levels be visualized in relation to geographic location and salary levels?
3. Location Opportunities: How can we build an interactive tool to help users find the highest-paying and nearest federal job posting based on the user's location preferences?

3 Methods

3.1 Accessing and Cleaning the Data

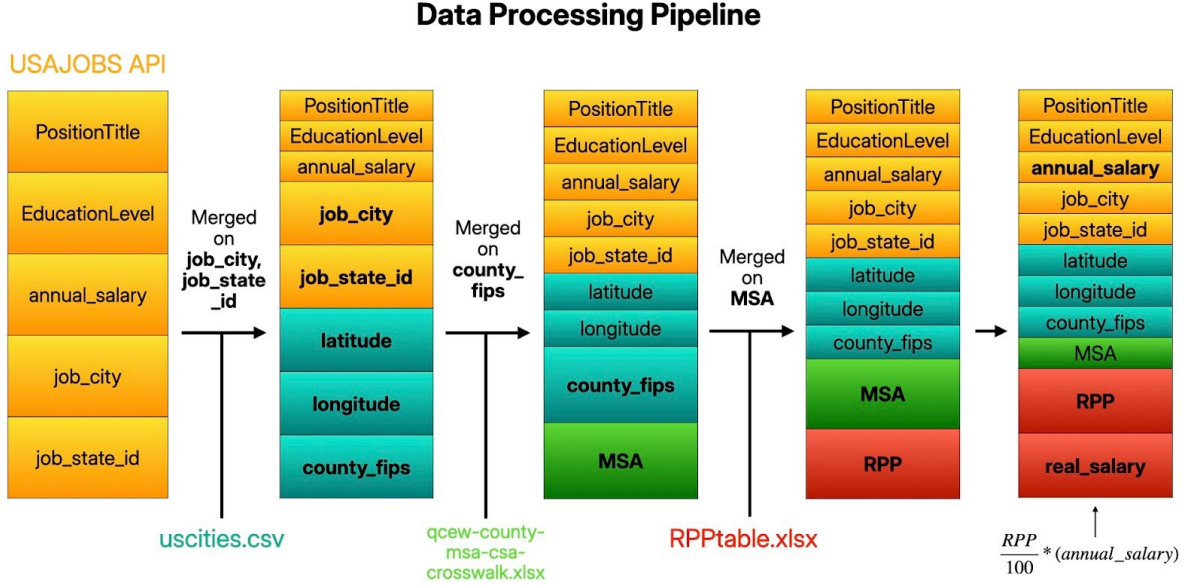


Figure 1: Data Cleaning Visualization

To begin understanding our data, we used an API key to retrieve all the jobs found through a search on USAJOBS. We iterated through each page, adding every job listing to a data frame. To start cleaning this large amount of data, we used a function, `clean_location2`, that uses two of the columns of raw location data to create a new column with cleaner syntax. It removes extra whitespace and commas, reformats the military bases and counties, separates states and cities, and normalizes the states into consistent cases and abbreviations. This is for easier location access for future applications and ensures that we pull all the possible data for the maps.

We imported other sets of data in order to analyze and contextualize the job listings. There were two general pieces of information that we wanted to add to each listing: latitude and longitude coordinates (for visualization) and price levels for the location of the listing. Initially, we attempted to use an API from Geocodio to geocode the location of each job listing into a pair of latitude and longitude values. However, Geocodio presented issues because it only allows for 2,500 free lookups per day, which was insufficient because the USAJOBS API contains over 8,000 listings and counting. Instead, we chose to merge in data from other external data sets.

The first of these was a CSV taken from SimpleMaps.com. This table contains all US city names, along with their corresponding FIPS codes and longitude and latitude values. This took care of our first goal, allowing us to create visual maps. Next, to address the objective of adding information on price levels, we imported an Excel spreadsheet from the U.S. Bureau of Labor Statistics containing the MSA (Metropolitan Statistical Area) code of each city. MSA codes are 5-digit numbers that identify geographical regions. The sole purpose of these MSA codes was to allow us to merge our data with another Excel

spreadsheet, which contains the RPP (Regional Price Parity) of every city. The RPP, expressed in %, is defined as the ratio of the price level in a given city in comparison with the average price level in the US.

With the new RPP data, we reformatted the data types and split the locations into a dictionary. Then, combining it with the cleaned data by key, we calculated the adjusted (real) salary and added it to the table. Throughout this whole process, we had many error checkpoints and ran various tests to ensure an effective manipulation of the data.

Our final data set consists of the original but cleaned data from USAJobs, including each job listing's title, education, and location, along with the nominal and real salary data for visual analysis.

3.2 Interactive Scatter Maps

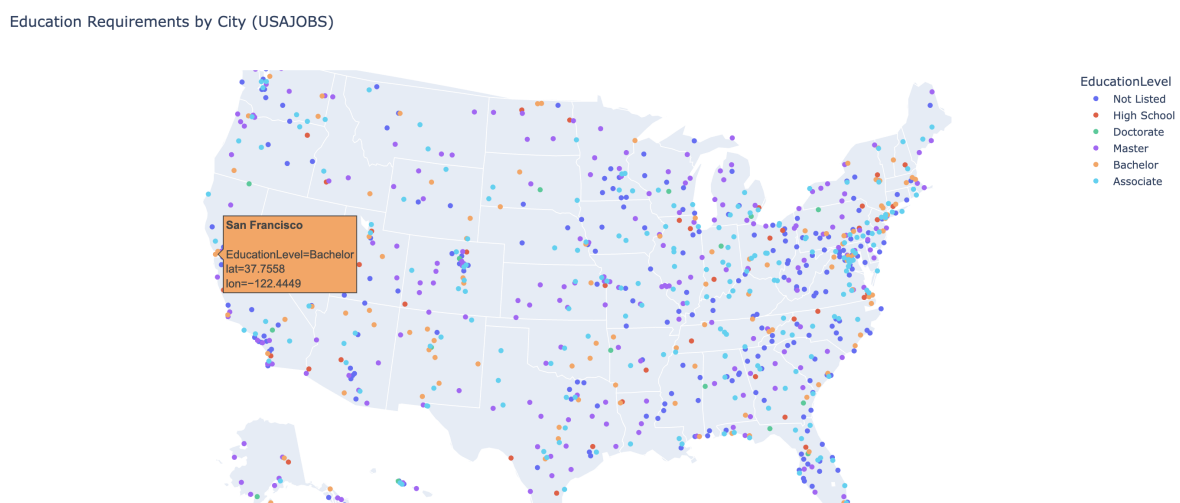


Figure 2: Interactive Scatter Map for Education Levels of Jobs by City

We sought to visualize the quality of jobs in each location. We decided that the best way to do this was a heat map since the results could be easily identifiable by hovering over the desired city. We started by creating a heat map to categorize our data by education level. We normalized the education levels under the qualification summary column to make the different names for the same level of education referenceable. Using this and the location data, we have an interactive map that shows each city's education level needed to get a job there (see Figure 2). However, all the jobs or dots were stacked on top of each other since we used the latitude and longitude values from cities' data to categorize them by location, causing the map to not be readable. To visualize this data, we instead averaged the salary per education level and created a table (see Figure 3).

Average Real Salary vs. Education Level	
Salaries are adjusted for Cost of Living (RPP)	
Education Level	Average Real Salary
Not Listed	\$125,742
Doctorate	\$107,918
Master	\$100,580
Associate	\$89,339
Bachelor	\$77,119
High School	\$47,767

Data based on USAJOBS API and BEA RPP data.

Figure 3: Education Level Table from USAJOBS Data

To use a map, we decided to count the number of jobs at each location and convert it into a geographical scatter plot (see Figure 4). Using a similar structure, we used the coordinates to count the amount of jobs at each location and visualized it by the size of the spot. This was a better application since the varying sizes of the location's dots also help the user understand how many jobs are at that location quickly, and it still utilized the locations in a clean way that showed all of our data.

USAJOBS Count by City

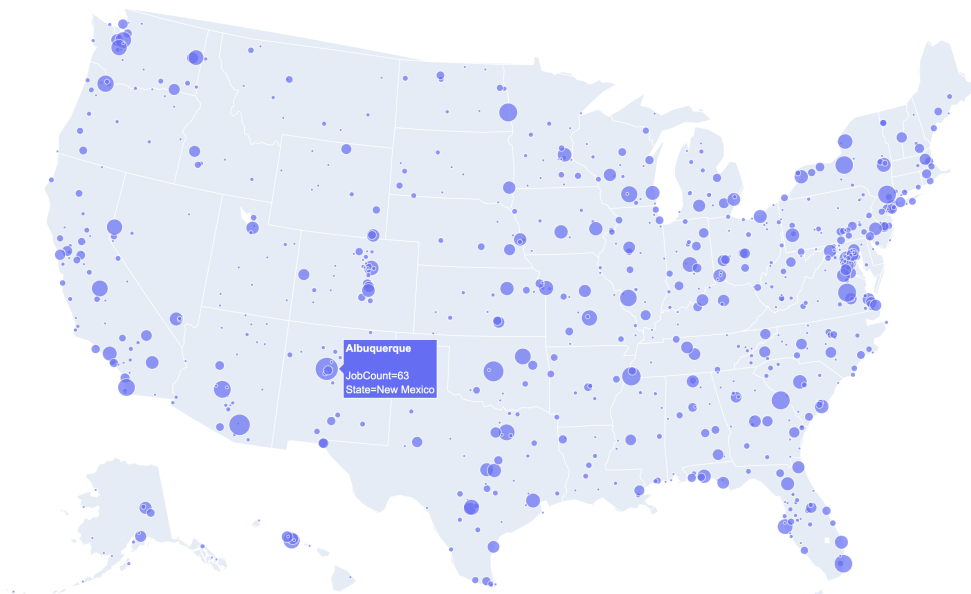


Figure 4: Interactive Scatter Map for Number Jobs by City

3.3 Job Search

Job Search Results Map



Figure 5: Job Search for Bachelor's Level Positions Near Davis, CA

On top of the larger pattern analysis, we created an interactive job search tool that will help users find federal jobs by using city coordinates and an adjustable distance limit, returning each job title, location, real salary, and distance from the specific location in miles (see Figure 5). We also added salary and education criteria to allow for greater customization and user utility. Users are able to filter jobs by education level, such as high school through PhD. Results can be sorted by highest real salary or nearest distance, all depending on the user's priorities. All of this was placed onto a map for easier readability, where job markers are sized and colored based on real salary, and the users' location is marked with a red star.

Several challenges came along with the development of this tool. Initially, the education filtering failed because the column was missing from our data. So, from here, it required regenerating the data set by running the pipeline that includes the education class function. The tool needed to work regardless of the users' education level, the distance limits, or the location preferences. It required checks to avoid errors when the parameters were "none," mainly for distant calculations where missing coordinates could crash the program.

3.4 Final Interactive Map

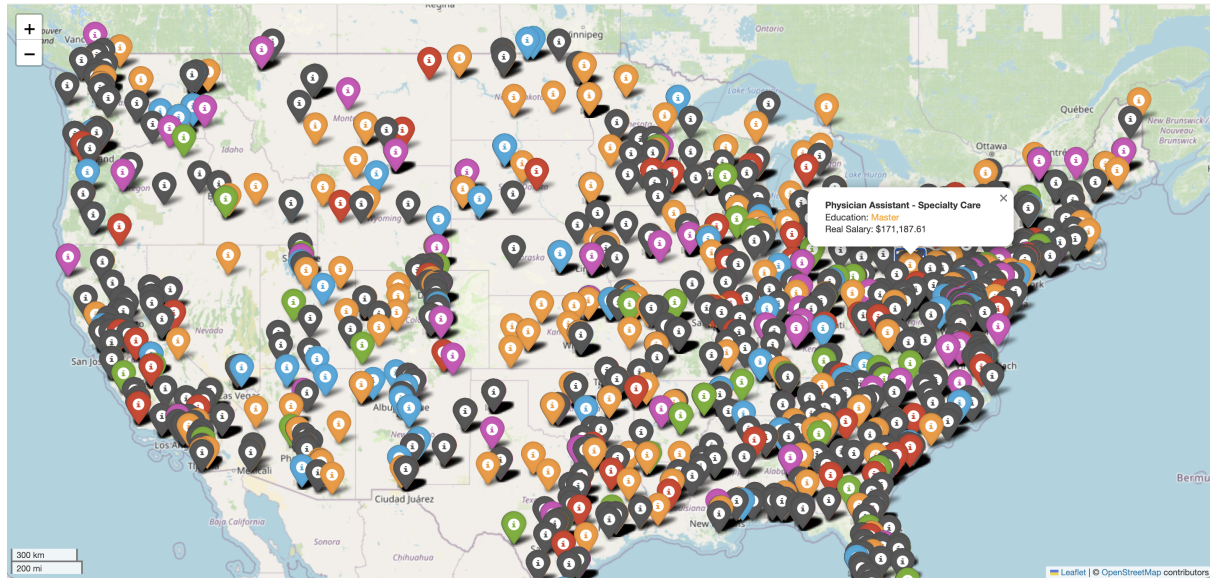


Figure 6: Final Interactive Map

To build off of our initial scatter maps (Figure 2, Figure 4), we decided to combine this information into one interactive map using the folium package (see Figure 6). Using the finalized data set consisting of title, education, real salary, and, most importantly, coordinates, we were able to loop through each job listing and mark them on an interactive JavaScript map using folium markers. Using a dictionary, each listing marker was appointed a different color (gray, green, purple, blue, orange, and red) based upon their respective education level required (education level not available, high school, associate's degree, bachelor's degree, master's degree, and PhD). Upon hovering over each marker, the position title is displayed. We also added an HTML/CSS division to display the position title (in bold), education level, and real salary upon clicking a given job listing's marker. Since our data only consists of job listings within the non-contiguous fifty states in the US, the folium map has boundaries to keep the user from straying away from North America.

4 Results

Looking at the heat map in Figure 2, we analyze patterns by geographical region. There is clearly a much higher density of jobs in the Eastern US compared to the West. States like Nevada, Montana, and Idaho have fewer than fifty jobs each, while most other states have one hundred or more listings. However, on the West Coast, there are still high levels of jobs in areas on the Pacific Coast. Cities like Tucson, San Diego, Portland, and Seattle have over thirty jobs each. The jobs around the eastern half appear to be pretty evenly divided, other than a large patch near the DC area, which makes sense since they are federal jobs. This gives a strong idea for trends across the larger US, allowing for someone to see where they may want to live or where more opportunities are if they want to get a government job and benefits. They can also zoom into their local area to analyze where the possible jobs are located near them and how many there are.

Looking at education levels (see Figure 3), we analyzed how they affect different salary levels. Using

the adjusted data to normalize salaries, we were able to see what someone would expect on average without a skew of geographical locations affecting the data. Most of the data makes sense, with a high school education being the lowest at \$47,949 per year. The highest education level is a Doctorate at \$108,199. The rest follow the pattern, with the least amount of years of education being the lowest pay, other than Associates earning higher than a Bachelor's by \$10,805 annually. The number of jobs that don't have a listed education level could be a possible reason for this conclusion, especially since it has the highest average salary. Associates are sometimes more specialized than a Bachelor's, which could also affect our data.

After building the interactive job search tool, Figure 4 shows a search for Bachelor's level jobs within 100 miles of Davis, California, sorted by highest real salary. The map shows 10 jobs found, with colored circles indicating salary. The results include jobs like Contract Specialist and Electronic Engineer positions at Defense Microelectronics in McClellan which is about 20 miles away with real salaries around \$126,413. There are also two positions in Sacramento, 15 miles away, paying between \$103,000 and \$112,000. Also, one job is actually in Davis, Social Services assistant paying about \$48,000. Salaries range from about \$45,000 to \$126,000 for Bachelors level in this area. The visualization helps users see both where jobs are located and how salaries compare, which primarily makes it easier to decide which opportunities to pursue based on the location and pay.

In contrast, Figure 5 is more of a visual tool rather than an interpretable visualization. Figure 5 allows one to get a general sense of the concentration of jobs, as distinguished by education level. For instance, as Figure 2 also suggests, Figure 5 shows that there is a greater concentration of job listings on the east side of the US, with significant variation in terms of education levels. Zooming in further, it can be observed that there are many jobs requiring master's degrees in Alaska and bachelor's degrees in New Mexico, as examples.

5 Summary

Answering our three goals, we were able to create tools and maps to visualize jobs in geographical regions and provide a tool for people to see job availability. We saw geographical patterns for federal jobs centered on the coasts, mostly the eastern half of the US, with very high density near the capital. To look at education, we used a table to compare different levels and their normalized salaries. Using the interactive maps, we are also able to search for jobs and salaries in one region, given distance and education requirements. Lastly, our final map (Figure 6) encompassed all of our questions. By showing all of the possible jobs on the map, color-coded for education, and containing their title and real salary information.

However, our project has limitations. The most obvious limitation is that our data only encompasses federal job listings. Our visualizations and analysis would be more practical if we included data spanning all sectors and industries, and included an interactive feature to also filter by sector. Additionally, the jobs are mapped by city, and their coordinates are that of the city centroid. Thus, if a user wanted to search for jobs within their own city, all of the listings would appear to be in the same location. In a large city such as Jacksonville, Florida, which spans over 700 square miles, this would not be very useful.

For the future, it would be interesting to track these values over time, using a time-series analysis. This would allow us to create better predictive models to help understand patterns of the federal job force. The data could also be turned into a very useful tool for the public to help them find jobs that fit

their needs and qualifications. With a competitive job market, finding jobs easier would be an asset to many people.

6 APA References

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- Note: This figure uses parameters: user_city= 'Davis', user_state= 'CA', max_results=15, sort_by= 'salary', max_distance =100, education_level='Bachelors'

6. https://github.com/ok-joe/STA-141B-Project/blob/98e5e0d0c2417bcab5a1437cd2db78106591256c/outputs/int_m

8 Code

All code can be found in this repository.